

第二次实验

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题目

请输入一个ASCII码整数 (65-90 or 97-122), 输出该ASCII码对应的字符。输入一个字符, 输出该字符的ASCII码。

设圆半径 $r=1.5$, 圆柱高 $h=3$, 求圆周长、圆面积、圆球表面积、圆球体积、圆柱体积。用cin输入数据, 输出计算结果, 输出时要求有文字说明。(课本p80, 2)

注: 圆球的表面积 $=4\pi r^2$, r 为球半径。

圆球的体积计算公式: $V=(4/3)\pi r^3$, r 为球半径

输入一个华氏温度, 要求输出摄氏温度。公式为 $c=5/9 \times (f-32)$, 输出要有文字说明。

运行结果

The screenshot shows the Visual Studio Code editor with a C++ file named `hw_1.cpp` open. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int ascii;
6     char character;
7
8     cout << "请输入一个ASCII码整数 (65-90 或 97-122): ";
9     cin >> ascii;
10    if ((ascii >= 65 && ascii <= 90) || (ascii >= 97 && ascii <= 122)) {
11        cout << "对应的字符是: " << char(ascii) << endl;
12    } else {
13        cout << "输入的ASCII码不在指定范围内。" << endl;
14    }
15
16    cout << "请输入一个字符: ";
17    cin >> character;
18    cout << "该字符的ASCII码是: " << int(character) << endl;
19
20    return 0;
21 }
```

The terminal window at the bottom shows the execution of the program:

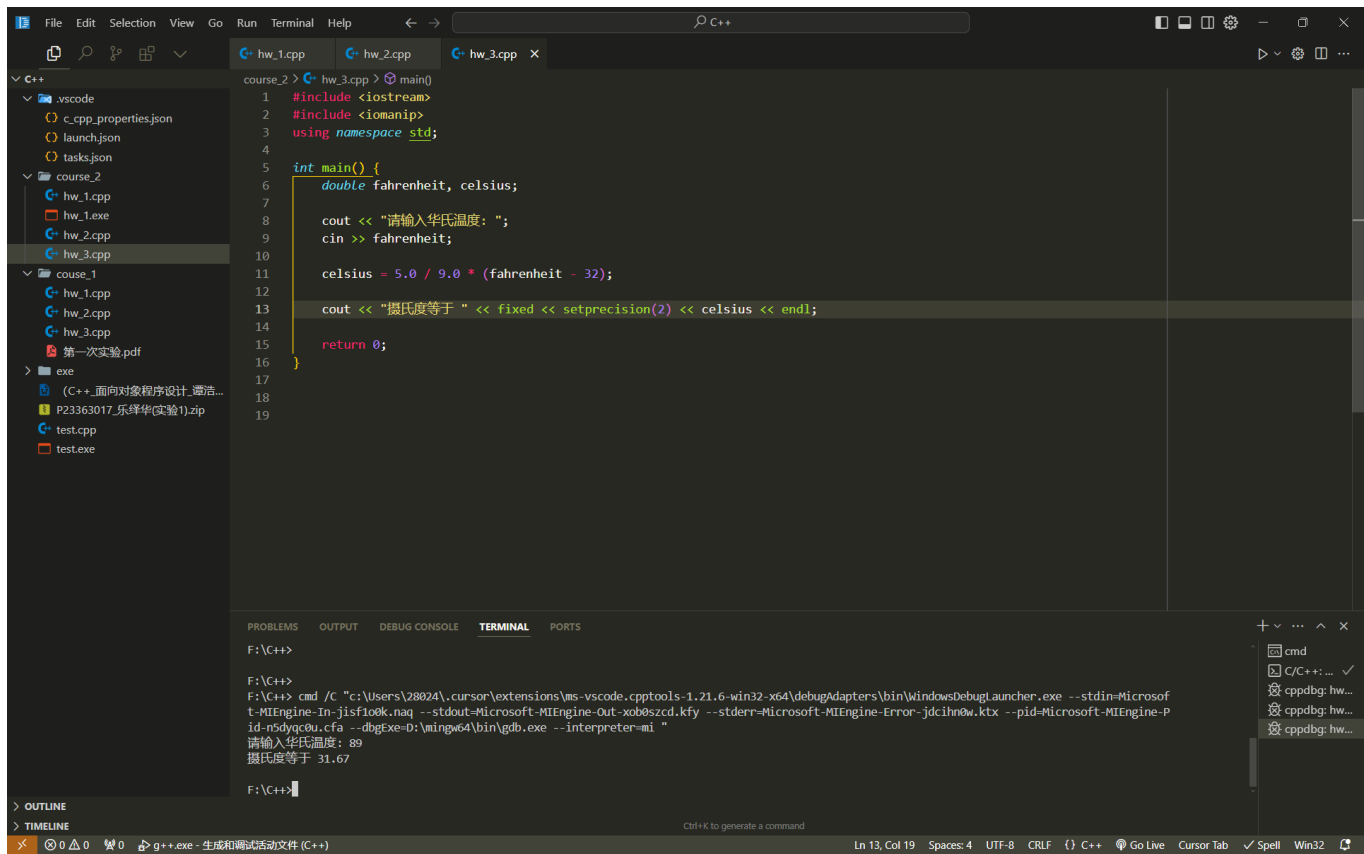
```
F:\C++>
F:\C++> cmd /C "c:\Users\28024\cursor\extensions\ms-vscode.cpptools-1.21.6-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe --stdin-Microsoft-MIEngine-In-hltxqmc.fl4 --stdout-Microsoft-MIEngine-Out-a55a3ph2.hhm --stderr-Microsoft-MIEngine-Error-z3dbbtbm.e2e --pid-Microsoft-MIEngine-Pid-uppn3uzz.n5x --dbgExe-D:\mingw64\bin\gdb.exe --interpreter=mi"
请输入一个ASCII码整数 (65-90 或 97-122): 66
对应的字符是: B
请输入一个字符: a
该字符的ASCII码是: 97
F:\C++>
```

The screenshot shows the Visual Studio Code editor with a C++ file named `hw_2.cpp` open. The code is as follows:

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 int main() {
6     const double PI = 3.14159;
7     double r, h;
8
9     cout << "请输入圆的半径: ";
10    cin >> r;
11    cout << "请输入圆柱的高: ";
12    cin >> h;
13
14    // 计算各种结果
15    double circumference = 2 * PI * r;
16    double circleArea = PI * r * r;
17    double sphereSurfaceArea = 4 * PI * r * r;
18    double sphereVolume = (4.0 / 3.0) * PI * pow(r, 3);
19    double cylinderVolume = PI * r * r * h;
20
21    // 输出结果
22    cout << "圆周长: " << circumference << endl;
23    cout << "圆面积: " << circleArea << endl;
24    cout << "圆球表面积: " << sphereSurfaceArea << endl;
25    cout << "圆球体积: " << sphereVolume << endl;
26    cout << "圆柱体积: " << cylinderVolume << endl;
27
28    return 0;
29 }
```

The terminal window at the bottom shows the execution of the program:

```
id-v5nkkagh.1yg --dbgExe-D:\mingw64\bin\gdb.exe --interpreter=mi"
请输入圆的半径: 1.5
请输入圆柱的高: 3
圆周长: 9.42477
圆面积: 7.06858
圆球表面积: 28.2743
圆球体积: 14.1372
圆柱体积: 21.2057
F:\C++>
```



总结与心得

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